

Impact of Virtual Self-Touch on the Embodiment of Avatars with Different Body Sizes

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ABSTRACT

This study explores the impact of our developed virtual self-touch (VST), where participants touch an avatar as a substitute body and receive haptic feedback, on updating the body schema for intervention in motor control strategies. VST was tested with ten participants using avatars of normal size and those with elongated forearms. The effectiveness of the method was evaluated using perceptual drift and reaching tasks, comparing VST to the arm swing task (AST). The results showed that VST enhanced perceptual drift and affected motor control strategy more than AST, suggesting its role in altering the body schema. The effect of this motor control strategy leads to rehabilitation.

Index Terms: Virtual Self-Touch, Embodiment, Haptic, Body Schema, Life and medical sciences, Virtual reality

1 INTRODUCTION

The body schema, a plastic framework for body recognition, adapts through sensory input and experiences, such as tool use, illusions, or virtual reality (VR). VR's ability to separate visual cues makes it a valuable tool for studying changes in the body schema, including adaptation to avatars with altered proportions. Techniques like the self-touch illusion (STI) [2] illustrate how synchronized sensory feedback can update the body schema. In contrast to building on Inamura et al.'s method[3] of using arm swing movements for avatar embodiment, this study introduces virtual self-touch (VST) in VR. In VST, participants interact with an avatar by virtually touching it, integrating visual, tactile, and proprioceptive sensations to facilitate updates to the body schema. The purpose of this study is to compare VST with previous approaches[3][5] using avatars with altered forearm lengths and thereby demonstrate its superior effectiveness in inducing perceptual drift and changing motor control strategies. These findings highlight VST's potential to enhance embodiment and its promising applications in rehabilitation.

2 METHODOLOGY

Participants experienced two types of avatars: Same-size avatars (SA), which matched their own body measurements, and enlarge avatars (EA), which had right forearms 1.5 times longer than the participants' right forearms. Two habituation methods were compared: arm swing task (AST), where participants moved the avatar's fingertip between virtual targets in time with a metronome, and VST, where participants touched each avatar fingertip with their own, receiving synchronized haptic feedback. In the experiment,

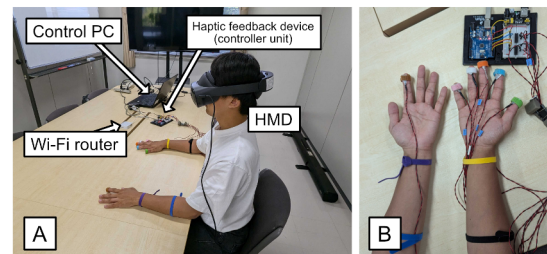


Figure 1: A: Experimental environment, B: vibrators on the tip of the fingers

we used a within-subject design: pretest evaluation, SA prequestionnaire, SA habituation (AST or VST), SA postquestionnaire, SA evaluation, switch to EA, EA prequestionnaire, EA habituation (AST or VST), EA postquestionnaire, and EA evaluation. Participants wore a meta quest pro HMD throughout, with avatar visibility toggled as needed. Habituation methods were tested on separate days to prevent carryover effects.

As shown in Fig. 1, the setup is as follows: a custom haptic feedback device with six fingertip vibrators controlled by an Arduino and PC; the meta quest pro HMD for VR display and upper-limb tracking; a unity VR application (running at 50 fps directly on the HMD) using the meta XR interaction SDK; custom avatars generated in Blender from human base meshes, exported in VRM format. A separate control application on a PC managed data logging (50 Hz), communication (Wi-Fi and USB), and haptic feedback control with approximately 17 ms delay from the avatar touch.

During habituation, participants sat with their forearms on a table, maintaining consistent arm postures (120° elbow, 10° shoulder abduction). Both habituation tasks lasted 5 min. Questionnaires assessing sense of body ownership (SoO) and sense of agency (SoA) using a seven-point Likert scale, as in a previous study[4], were administered before and after each habituation session. The forearm angle relative to the trunk's frontal plane was controlled.

The evaluation comprised questionnaires (about SoO and SoA), perceptual drift measurement (pointing to their own fingertip without avatar visibility, with normalized elbow-fingertip distance measured (Fig.2A)), and a reaching task (reaching for a target while avoiding cylindrical obstacles, also without avatar visibility except to period of initial position adjustment). As shown in Fig. 2B, the key metrics in the reaching task include average clearance (AC), average fingertip-obstacle distance during avoidance (between 45° and 90° fingertip position angle, with the start position being 0°); reaching rate X (RR_x) and reaching rate Y (RR_y), the x- and y-axis errors between the final fingertip position and the target, averaged over the last ten data points. The reaching task data were transformed into a planar coordinate system, temporally normal-

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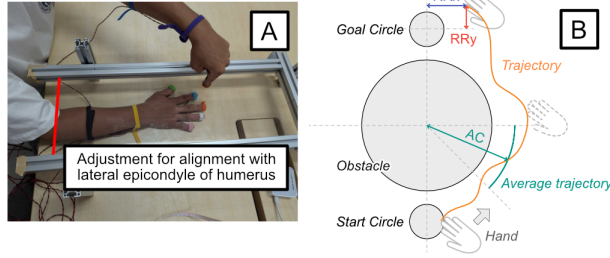


Figure 2: A: Measuring instruments for forearm length, B: outline of reaching task evaluation metrics and coordinate definition of evaluation space

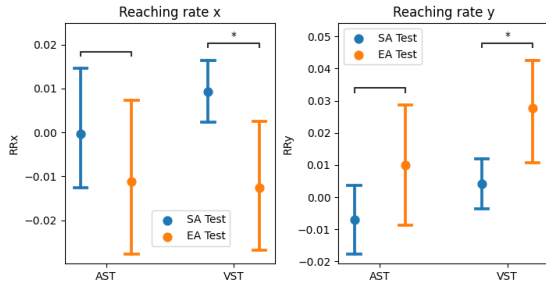


Figure 3: Reaching rates (RR_x and RR_y) for AST and VST, * means p-value < 0.05. $n = 9$.

ized (100 interpolated points using cubic spline), and normalized for the pretest value.

In the statistical analysis, we used the Wilcoxon signed-rank test ($\alpha = 0.05$), with Bonferroni correction for multiple comparisons. Ten healthy adults (eight males and two females, 23.6 ± 1.7 years old) participate. The experiment was conducted with the approval of the Ethics Committee for Research Involving Human Subjects at Graduate School of Engineering Science, Osaka University(R3-17-2).

3 RESULTS

In the questionnaires, AST showed significant differences ($p < 0.05$, Wilcoxon signed-rank test, $n = 10$) in both SoA and SoO pre/post for SA and EA. VST showed a significant difference only in the SoO post-EA. The mean and standard deviation (SD) of SoA and SoO scores in AST are as follows: pre-SA SoA 5.5 ± 0.8 , SoO 4.6 ± 1.3 ; post-SA SoA 6.2 ± 0.6 , SoO 5.3 ± 1.2 ; pre-EA SoA 5.5 ± 1.1 , SoO 4.6 ± 1.2 ; post-EA SoA 5.9 ± 0.7 , SoO 5.3 ± 1.4 ; the mean and SD in VST are as follows: pre-SA SoA 5.2 ± 1.1 , SoO 4.4 ± 1.2 ; post-SA SoA 4.4 ± 1.6 , SoO 4.6 ± 1.8 ; pre-EA SoA 4.7 ± 1.5 , SoO 4.3 ± 1.3 ; post-EA SoA 4.8 ± 1.6 , SoO 5.0 ± 1.3 .

Perceptual drift showed significant differences ($p < 0.0167$, Bonferroni corrected Wilcoxon signed-rank test, $n = 10$) between SA/EA and pretest/EA in AST and VST. The mean and SD of perceived forearm length in AST are as follows: pretest 0.82 ± 0.10 , SA 0.83 ± 0.12 , EA 0.86 ± 0.12 ; the mean and SD in VST are as follows: pretest 0.83 ± 0.10 , SA 0.84 ± 0.11 , EA 0.91 ± 0.10 .

The reaching task results ($n = 9$ due to data loss) showed significant AC differences ($p < 0.05$, Wilcoxon signed-rank test) between SA/EA in AST and VST. The mean and SD of AC in AST are as follows: SA 0.0037 ± 0.0123 , EA -0.0091 ± 0.0181 ; the mean and SD in VST are as follows: SA 0.0112 ± 0.0167 , EA -0.0113 ± 0.0280 . RR_x and RR_y showed significant differences only in VST between SA/EA (Fig.3).

4 DISCUSSION

In AST, both SoA and SoO significantly improved, indicating enhanced embodiment. SoA was influenced by avatar size fidelity, whereas SoO was less influenced. In VST, only EA SoO significantly improved. The voluntary movement in AST may explain the SoA improvement, which is absent in VST. Sensor accuracy issues in VST (fingertip contact misalignment) likely hindered SoO improvement, especially in SA, where physical contact with the participant's hand was more likely. The EA condition of VST showed SoO improvements comparable to the EA condition of AST, indicating the potential of VST when physical contact is avoided.

Perceptual drift showed significant differences between pretest/EA and SA/EA in AST and VST, indicating drift in the elongation direction. No significant difference was found between pretest and SA. EA drift was greater in VST (0.08) than in AST (0.04). The greater drift in VST may be attributed to multisensory integration (visual and tactile).

In the reaching task, AC significantly decreased between SA/EA in AST and VST, indicating reduced clearance. RR_x and RR_y showed significant differences only in VST, with RR_x decreasing and RR_y increasing, indicating a shift to the upper left. This suggests that participants may use different spatial coordinate systems for proximal (relative forearm/object size) and distal (tactile reach distance) space. This study differs from previous study in that VST also applies feedback to the "not only touched finger but also touching one" and shows the possibility of intervening in motor control strategy[1].

5 CONCLUSION

This study compared VST (haptic feedback during avatar self-touch) with AST for inducing avatar embodiment, revealing distinct effects on body schema. VST improved SoO with EA avatars but was less effective with SA avatars, which could be attributed to conflicting sensory information. VST also induced larger perceptual drift. The reaching task results suggest the possibility of intervening in motor control strategies. However, these results are based on a very small sample size and need to be validated with a larger sample size. Further experiments should randomize the protocol and improve the discrepancy between the real and VR environments.

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